PH.45320 **FALL 2026**

Static Cosmology

Lecture 5: Equilibrium Helium Abundance

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Overview

Helium abundance is obtained as a stable fixed point of production, destruction, and diffusion terms.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$dY_i/dt = P_i(Y) - D_i(Y) + \nabla \cdot (D\nabla Y_i)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$P_i(Y_*) = D_i(Y_*)$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\tau_{relax} = 1/|Re\lambda_{min}|$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 5
- Equilibrium Composition Table EC-14
- Nuclear block NB-19